

Ariel Fu

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Authorized to work for any employer with no restrictions

EDUCATION

Northeastern University

Boston, MA

Master of Science in Computer Science GPA: 3.67

Expected graduation: Dec 2026

Courses: Fundamentals of Computer Networking, Fundamentals of Cloud Computing, Object-Oriented Design, Database Management, Mobile Application, Foundation of AI

SKILLS

Programming Languages: Python, TypeScript, JavaScript, Swift, SQL, Bash, HTML, CSS

Frameworks/Libraries: React, React Native, Expo, Node.js, Express, SwiftUI, REST APIs

Cloud/Data: AWS, Terraform (IaC), GitHub Actions (CI/CD), MongoDB, SQL, Firebase, Linux, Git

PROFESSIONAL EXPERIENCE

Software Engineer, Internship

Feb 2025 – Apr 2025

A Round Entertainment

Remote, United States

- Developed and maintained an iOS mobile application using **React Native, Expo, and TypeScript**, building reusable component hierarchies that improved code maintainability and reduced UI duplication by 30%
- Designed and integrated **RESTful APIs with Node.js** and Express to support real-time messaging workflows, ensuring reliable client-server communication and structured data exchange.
- Modeled and queried application data using **MongoDB and SQL** for structured data analysis, improving data retrieval efficiency and supporting backend feature development
- Implemented structured **state management** patterns and optimized frontend-backend synchronization, reducing unnecessary re-renders by 25% and improving UI responsiveness
- Collaborated using **Git-based workflows** including feature branching, pull requests, and peer code reviews, reducing integration conflicts and improving deployment efficiency

PROJECTS

Production Three-Tier AWS Architecture | *AWS, Terraform, EC2, RDS, ALB*

Aug 2026 – Sep 2026

- Designed and provisioned a highly available **three-tier AWS architecture** across multiple Availability Zones using Terraform, including VPCs, public/private subnets, routing, and NAT gateways
- Deployed scalable application infrastructure using **EC2, Application Load Balancer, and Auto Scaling**, enabling automated health checks, traffic distribution, and instance recovery
- Secured the application and database tiers using private subnets, security groups, IAM least-privilege roles, Amazon RDS, and AWS Secrets Manager for credential management
- Automated infrastructure deployment with **GitHub Actions and OIDC-based AWS authentication**, and implemented CloudWatch monitoring, Terraform testing, and CI/CD validation

Cloud Resume Challenge | *AWS, Terraform, GitHub Actions, Python, JavaScript*

Jun 2026 – Jul 2026

- Built and deployed a serverless AWS web application using **CloudFront, S3, API Gateway, Lambda, and DynamoDB**, serving a static frontend with a dynamic visitor counter
- Provisioned and managed cloud infrastructure using **Terraform**, implementing **Infrastructure as Code (IaC)** with remote state stored in Amazon S3.
- Designed and automated a **GitHub Actions CI/CD pipeline** to validate Terraform, generate deployment plans, apply infrastructure changes, and invalidate CloudFront cache after deployment
- Implemented secure **OIDC authentication** between GitHub Actions and AWS IAM, eliminating long-lived AWS credentials through least-privilege role assumption

Serverless Café Sales Reporting System | *Python, AWS Lambda, RDS*

May 2026 – Jun 2026

- Built a serverless daily sales reporting workflow using **AWS Lambda**, Amazon RDS, SNS, EventBridge Scheduler, Secrets Manager, IAM, and CloudWatch.
- Configured **VPC** networking and security groups to allow a Lambda function in private subnets to securely access an RDS database.
- Troubleshooted **IAM** permissions, Lambda handlers, database connectivity, and scheduled executions using CloudWatch Logs.